Notes on Radio Evolution of Galaxies

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Contents

Chapter 1

Useful Terminology and Concepts

TODO

- need to add Initial Mass Function stuff at some point.
- need to add info about emission lines of x rays (detecting hot gas $\sim 10^7$ K), Hα lines (detecting warm gas $\sim 10^4$ K), HI (21 cm) lines (detecting cold gas ~ 100 K).
- From sec 9.6 in Schneider, need to add info about star formation detection, IMF and other stuff. Whole treasure trove is there.
- X-factor conversion for estimation of H_2 abundance given we know CO abundance.

1.1 Basic Terminology

need to add magnitude concise defns, colour index followed.

i. add parsec defn

ii. **Luminosity** (L): The total amount of electromagnetic **energy emitted per unit time** by an astronomical object. The total luminosity from an isotropic source is:

$$L = \int \mathrm{d}A\, I(R) = \int_0^{2\pi} \int_0^{\infty} \mathrm{d}\theta\, \mathrm{d}R\, I(R)\, R$$

iii. **Radiant Flux Density** (S): The total amount of electromagnetic **energy per unit area per unit time** by an astronomical object. In other words, it is the luminosity per unit area. According to this definition, at a distance D from a star (say):

$$S = \frac{L}{4\pi D^2}$$

iv. **Effective Temperature** (T_{eff}): The effective temperature of a body such as a star or planet is the temperature of a black body of the same radius that would emit the same total energy as electromagnetic radiation.

v. **Effective Radius** (R_{e}): Defined such that half of the luminosity of a galaxy is emitted from within this radius.

$$\int_0^{R_e} \mathrm{d}R\ I(R)R = \frac{1}{2}\int_0^{\infty} \mathrm{d}R\ I(R)R$$

vi. **Effective Intensity** (I_e): $I_e = I(R_e)$; Total intensity from within the effective radius.

vii. **Colour Temperature** (T_c): The temperature of a black body such that it has the same colour index as that of an astral body.

viii. **Specific Luminosity** (L_ν): The energy emitted per unit time for a unit frequency range.

ix. **Rotation Curves**: Rotational velocity (v) of stars and gas around the galactic center as a function the galacto-centric radius[1] R.

x. **Globular Clusters**: A globular cluster is defined as a gravitationally bound collection of typically several hundred thousand stars contained within a spherical region with a radius of approximately 20 parsecs (pc). The members of the cluster orbit according to the gravitational potential of the cluster while the cluster as a whole orbits the center of the galaxy.

xi. add sun units, $\odot$

Spiral Galaxy Specific

i. **Scale-length** (h_R): The radius at which the surface brightness of the galaxy is a factor of e less than at the center.

ii. **Scale-height** (h_z): The radius at which the surface brightness of the galaxy (edge-on) is a factor of e less than at the center.

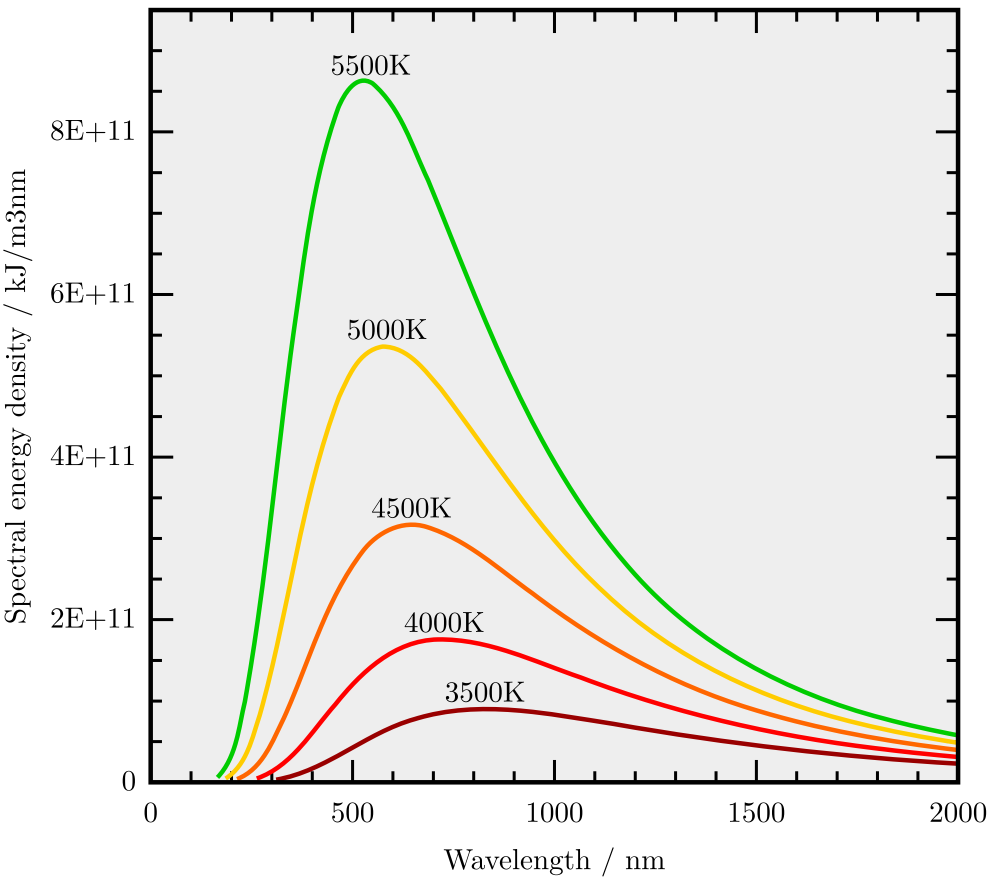
[1] Distance from the center of the galaxy.

1.2 Blackbody Radiation

A very brief explanation of blackbody radiation includes the Stefan-Boltzmann Law which states that the flux S for a perfect blackbody is given by:

$$S = \sigma T^4 \tag{1.1}$$

where $\sigma = 5.67 \times 10^{-5}$ erg cm^{-2} K^{-4} s^{-1} is the Stefan-Boltzmann constant.



Additionally, for a blackbody, it helps to know the Wien's Displacement Law which states that the emission curve for blackbodies peak at different wavelengths which is inversely proportional to its temperature:

$$\lambda_{\text{peak}} = \frac{b}{T} \tag{1.2}$$

where $b \approx 2.898 \times 10^6$ nm K.

1.3 The Magnitude Scale

Our eyes judge brightness according in a logarithmic fashion, similar to hearing. This means that what we perceive to be twice as bright is actually emitting a lot more than just twice the radiant energy. Due to this, optical astronomy uses 'magnitude' as a scale of brightness.

1.3.1 Apparent Magnitude

The purpose of apparent magnitude is to closely match visually reported magnitudes and quantify these historical observations. It is a measure for the radiant flux S from a body. For two sources with fluxes S_1 and S_2, their *apparent magnitudes* m_1 and m_2 are defined as:

$$m_1 - m_2 = -2.5 \log\left(\frac{S_1}{S_2}\right)$$

This means that a brighter object will have a smaller apparent magnitude. The factor of 2.5 is chosen to most accurately match the historically reported magnitudes. A more *absolute* definition can be found below in the next section, which depends on the filter in consideration.

1.3.2 Filters and Colour

Optical observations are performed with a combination of filters and detectors. Since the spectral energy distribution is not necessarily uniform over all wavelengths, apparent magnitudes are defined for each of these filters, i.e. apparent magnitude is a function of the wavelength of light being considered.

Vega Magnitude

$$m = m(\lambda) \equiv m_X \equiv X$$

where X is the filter for some specific wavelength λ.

The magnitudes of the different filters are related, by definition, as $U = B = V = R = I = \dots$ for main sequence stars of spectral type A0 (Vega is an archetype of such stars), i.e., every colour index for such a star is defined to be zero.

For a more precise definition, let $T_X(\nu)$ be the transmission curve of the filter-detector system. $T_X(\nu)$ specifies which fraction of the incoming photons with frequency are registered by the detector. The apparent magnitude of a source with spectral flux S_ν is then:

$$m_X = -2.5 \log\left(\frac{\int \mathrm{d}\nu\, T_X(\nu)\, S_\nu}{\int \mathrm{d}\nu\, T_X(\nu)}\right) + \text{const.}$$

The argument of the logarithm is the average flux received in total for the entire EM spectrum. The constant is determined from reference stars. This is also referred to as the Vega apparent magnitude.

AB Magnitude

Another commonly used definition, does not consider the spectral energy distribution of reference. The reference is instead defined as constant flux at all frequencies, $S_\nu^{\text{ref}} = S_\nu^{\text{AB}} = 2.89 \times 10^{-21}$ erg s^{-1} cm^{-2} Hz^{-1}. This value makes A0 stars like Vega have the same magnitude in the original Johnson-V-band as in the AB system.

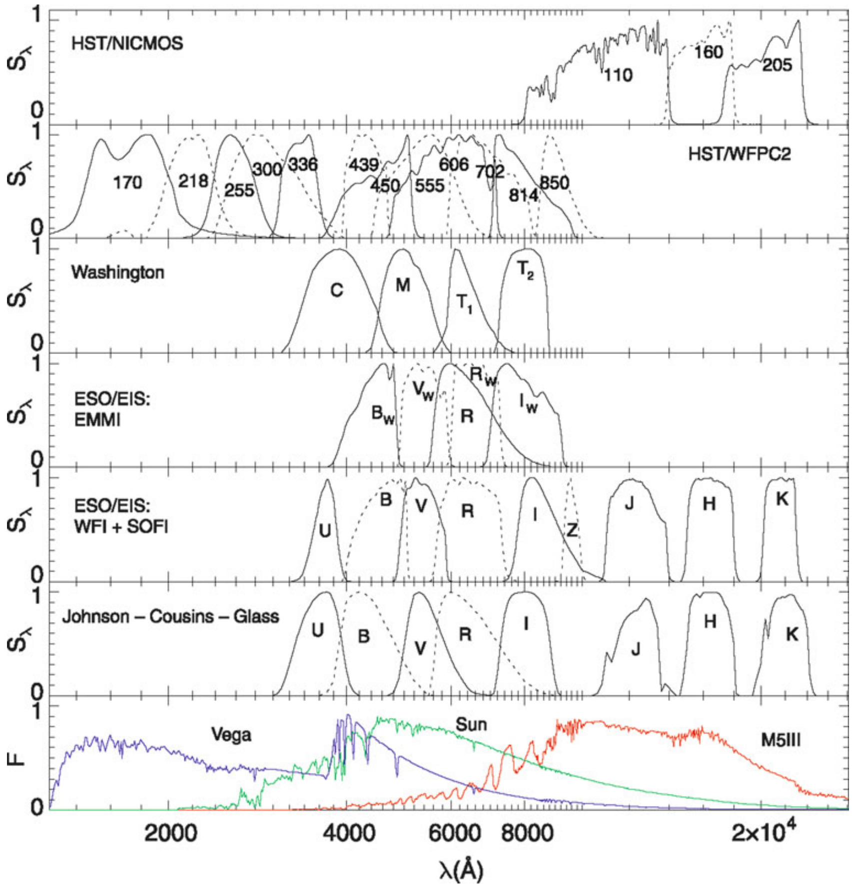


Figure 1.1: This shows the transmission curves of different spectra. At the bottom, we have the spectra of 3 stars with different effective temperatures.

1.3.3 Absolute Magnitude

Since apparent magnitude does not reveal any information about the luminosity of the body since it inherently depends on the distance of the body from Earth. If we are at a distance D from a body and L_ν (energy per unit time for unit frequency range) is its specific luminosity, the flux for isotropic emission is given by:

$$S_\nu = \frac{L_\nu}{4\pi D^2} \tag{1.3}$$

To reveal the physical properties of the source itself irrespective of how it appears to us, we define absolute magnitude, M_X, where X refers to the filter under consideration. It is defined to be the apparent magnitude of a body at a distance of 10 pc from us. Its relation to the apparent magnitude is as follows:

$$m_X - M_X = 5\log\left(\frac{D}{1pc}\right) - 5 \equiv \mu$$

where μ is called the *distance modulus*. Hence, the latter is a logarithmic measure of the distance of a source: $\mu = 0$ for $D = 10$ pc, $\mu = 10$ for $D = 1$ kpc, and $\mu = 25$ for $D = 1$ Mpc. The difference between apparent and absolute magnitude is independent of the filter choice, and it equals the distance modulus if no extinction is present. In general, this difference is modified by the wavelength (and thus filter) dependent extinction coefficient.

1.3.4 Bolometric Parameters

The total flux S and luminosity L over all frequencies are naturally be defined as:

$$S = \int_0^\infty S_\nu \,\mathrm{d}\nu; \qquad L = \int_0^\infty L_\nu \,\mathrm{d}\nu$$

These define the *bolometric magnitudes*:

$$m_{\mathrm{bol}} = -2.5\log(S) + \mathrm{const.}$$
$$M_{\mathrm{bol}} = -2.5\log(L) + \mathrm{const.}$$

The constants can be determined from reference stars. In particular, using the sun as reference with $m_{\odot\mathrm{bol}} = -26.83$ and the distance of 1 a.u. gives $\mu = -31.47$, thus giving us:

$$M_{\odot\mathrm{bol}} = m_{\odot\mathrm{bol}} - \mu = 4.74$$

From this, we obtain:

$$M_{\mathrm{bol}} = 4.74 - 2.5\log\left(\frac{L}{L_\odot}\right)$$

where the luminosity of the sun is $L_\odot = 3.85 \times 10^{33}$ erg s^{-1}. To obtain a value for the luminosity for galaxies or AGNs[2], we use the relative definition of absolute magnitude to write:

$$L_X = 10^{-0.4(M_X - M_{\odot X})} L_{\odot X} \tag{1.4}$$

where X represents the X-band filter.

1.4 Stars

Stars can be reasonably thought of as gas spheres, in the cores of which light atomic nuclei fuse into heavier ones (mostly hydrogen to helium) by thermonuclear processes, thereby producing energy. **External appearance of stars** is primarily determined by their radii R and their temperature T. **Properties and evolution of stars** is primarily governed by their masses M.

1.4.1 Parameters of Stars

For a first approximation, the spectral energy distribution of a star's emission can be described by blackbody radiation, where the flux S is given by the Stefan-Boltzmann equation (1.1).

[2]Active Galactic Nuclei (AGN) are galaxies whose largest fraction of luminosity is produced in a very small central region: the active galactic nucleus.

Thus, the luminosity for the star at a distance R is given by (1.3):

$$L = 4\pi R^2 \ \sigma T^4$$

Stars deviate from this profile and we define effective temperature (iv.) as:

$$\sigma T_{\text{eff}}^4 \equiv \frac{L}{4\pi R^2} \tag{1.5}$$

The luminosities can range from $10^{-4} L_\odot$ to $10^5 L_\odot$.

The temperature of a star can be estimated from its colour. Again, to a first approximation, Wien's displacement law (1.2) can explain the colour of stars depending upon temperature. mention how redshift gets factored in here, found relevant thread here.

Using the colour index $X - Y \equiv m_X - m_Y$ in two filters X and Y lets us find the colour temperature T_c (vii.) of the star. For a perfect blackbody, $T_c = T_{\text{eff}}$ but in general they differ.

1.4.2 Classification of Stars

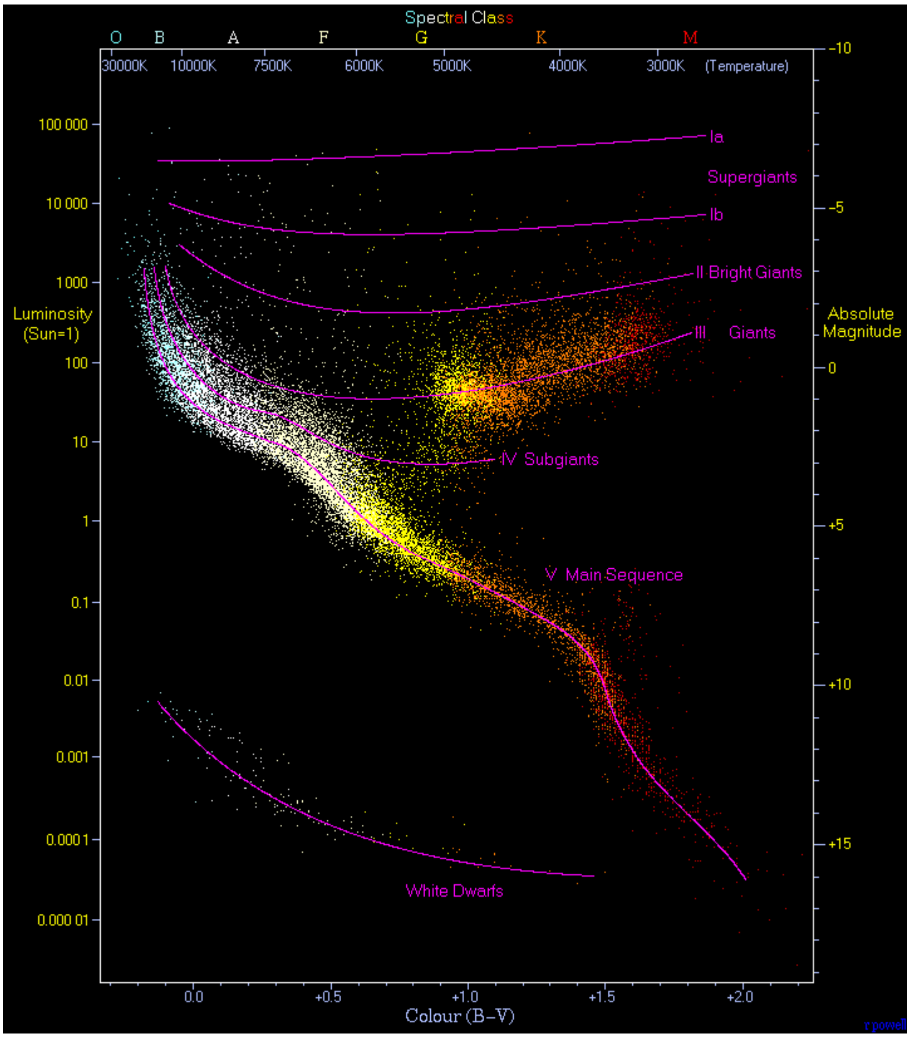
The Harvard sequence of stellar spectra classifies stars according to the **widths and ratios of their atomic spectral lines**. This is essentially a classification based on **colour temperature** T_c since the transitions leading to those spectral lines are directly dependent on the temperature of the region.

The sequence of **spectral classes** is denoted in the order O, B, A, F, G, K, M in decreasing order of colour temperature. Each spectral class is further subclassified using digits 0 to 9, again in descending order of colour temperature. For instance, our Sun is a G2 star.

Hertzsprung-Russell Diagram (HRD): Absolute magnitude vs. spectral classes.

Colour-Magnitude Diagram (CMD): Absolute magnitude vs. colour index (like $B-V$ or $V-I$)

Both of these diagrams yield a very charcteristic central result of astronomy which says that most stars are arranged along a 1-dimensional sequence - the main sequence. Luminosity L vs effective temperature T_{eff} also yields a different but very similar graph.



Based on luminosity, the stars can be classified into **luminosity classes**:

Class I Supergiants; Class II Bright giants

Class III Giants; Class IV Subgiants

Class V Dwarfs; Class VI Subdwarfs

Our sun is a luminosity class V dwarf.
Since stars exist which have, for the same spectral type and hence the same color temperature (and roughly the same effective temperature), very different luminosities, we can deduce immediately that these stars have different radii from the equation of effective temperature (1.5). Thus the red giants which have the same effective temperatures as the main sequence stars must have much larger radii.

Spectroscopically, the width of spectral absorption lines depends on gravitational acceleration on star's surface, $g = \frac{GM}{R^2}$, according to models of stellar atmosphere. It is based on these line widths (and thus luminosity) that we obtain the different luminosity classes.

If we have the distance of a star (thus its luminosity) and if its surface gravity can be derived from the line width, we can obtain the **stellar mass**. For the **main sequence stars**, this turns out to be approximately:

$$\frac{L}{L_\odot} \approx \left(\frac{M}{M_\odot}\right)^{3.5} \tag{1.6}$$

Sun

The solar spectrum peaks at around 501 nm, which looks green ish to me. According to Wien's law, assuming blackbody, the surface temperature of the sun is estimated to be:

$$T_{\text{eff}} = \frac{b}{501 \text{ nm}} \approx 5784.431\text{K}$$

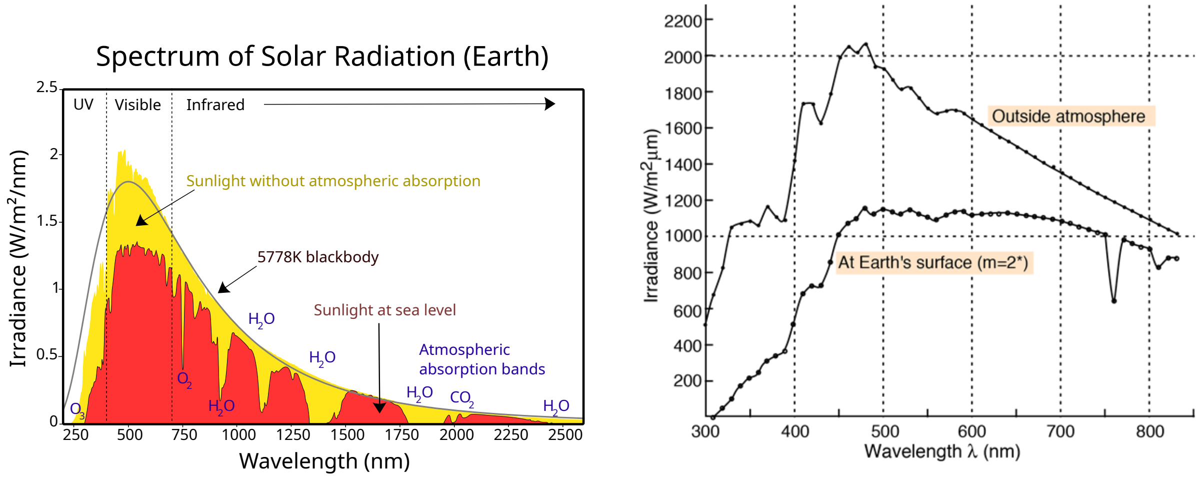


Figure 1.2: Solar Spectrum, G2 Type Star

1.4.3 Spectra of Some Stars

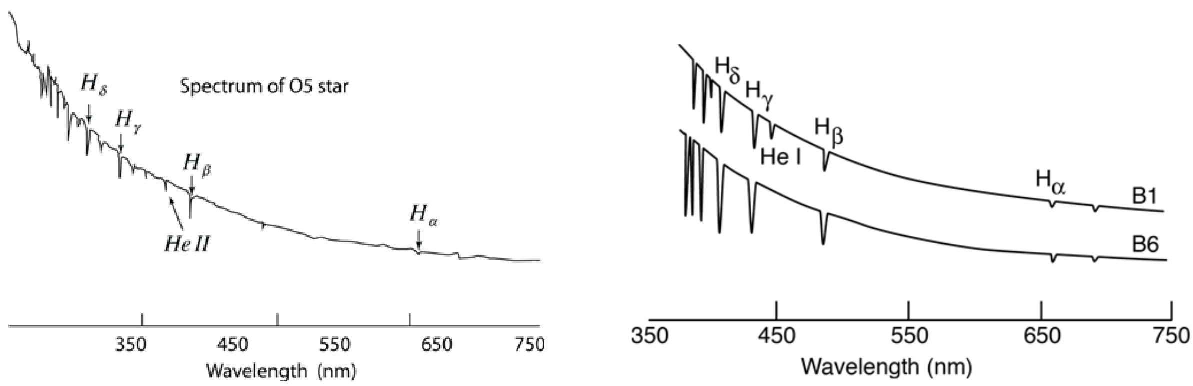


Figure 1.3: Spectrum of an O5 Type Star (left) and B Type Stars (right).

The different spectral types have different effective temperatures. This is demonstrated in their emission spectra where the earlier stars have a smaller peak wavelength, thus appearing bluer.

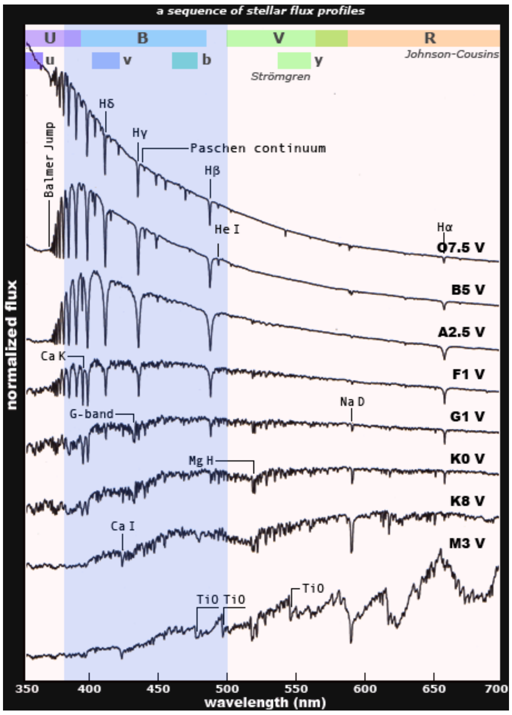


Figure 1.4: The spectra of the different spectral classes of stars.

1.4.4 Structure and Evolution of Stars

Add how stars are formed from cool gas through gravitational forces mainly.

Stars are almost spherically symmetric, so we consider its parameters as radial functions: Pressure, Density, Temperature and Chemical Composition. Stars are almost always in hydrostatic equilibrium, with the pressure fighting against its own gravitational pull.

Fusion at the Center and Transport of Energy

In the center of a star, the density and temperature are high enough to cause thermonuclear ignition, causing the formation of Helium from Hydrogen (4 protons combine into ^{4}He nucleus). This fusion results in energy release that is transported outwards in the form of radiation transport or through convection of stellar plasma. It is then released in the form of electromagnetic waves from the star. The chemical composition is homogenous in the convection zones as they transport and mix the stellar material.

A part of the energy released is in form of neutrinos that can escape unobstructed from stars because of their low cross-section[3].

Temperature Dependence of Energy Production Rate

For temperatures below around 15×10^6 K, the energy production rate is approximately proportional to T^4, where the reaction follows the 'pp-chain'. Above that, the reaction follows the 'CNO cycle' where the rate is roughly proportional to T^{20}.

Lifetime of a Star

Beginning with homogenous chemical composition, if the stars have masses above $\sim 0.08 M_\odot$, only then do they have enough temperature and pressure in their core to fuse hydrogen into helium. Otherwise, they are called brown dwarfs which aren't exactly stars in the proper sense.

[3] Cross-section is a measure related to the probability that a specific process will take place in a collision of two particles. In this case, it is referring to the probability that a neutrino will interact with matter.

At the onset of nuclear fusion, the star is located on the Zero-Age Main Sequence (ZAMS) in the HRD. With the conversion of hydrogen into helium, the chemical composition changes and the conditions in a star will change considerably once roughly 10% of its hydrogen is used up. At that point, the stars will move away from the main sequence. This allows us to estimate the life of a star on the main sequence. The total energy produced in this phase is:

$$E_{\rm MS} = 0.1 \times M\ c^2 \times 0.007$$

where $M\ c^2$ is the rest mass of the star and 0.007 is the efficiency of the conversion. We know that luminosity is energy per unit time, so this energy is equivalent to the product of the luminosity in this period with the time period $t_{\rm MS}$:

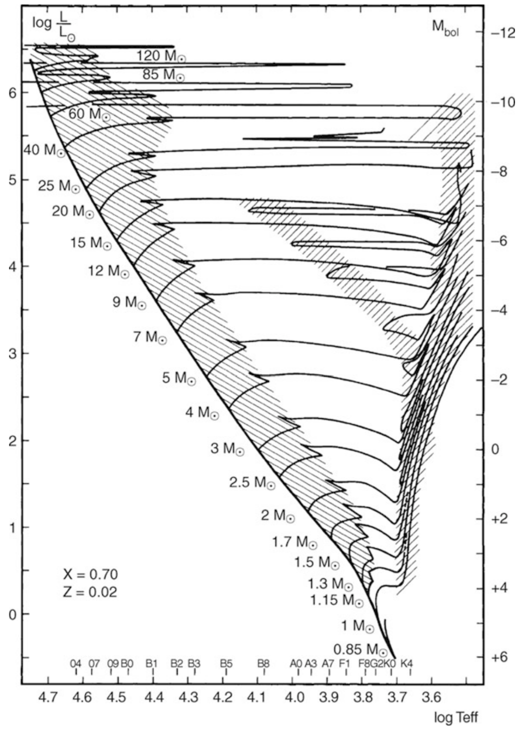


Figure 1.5: The solid line represents the ZAMS in this temperature-luminosity diagram.

$$t_{\rm MS} = \frac{E_{\rm MS}}{L} \approx 8 \times 10^9\ \frac{M/M_\odot}{L/L_\odot}\ {\rm yr} \approx 8 \times 10^9 \left(\frac{M}{M_\odot}\right)^{-2.5}\ {\rm yr}$$

where we used the fact that the luminosity is a steep function of the stellar mass (1.6).

Thus, heavier the star, lesser the time it spends on the main sequence. A G star like our sun stays in the main sequence for about 8-10 billion years, whereas an O or B star remains only for a few million years. In the course of their evolution on the main sequence, the stars move away slightly from the ZAMS in the HRD towards somewhat higher luminosities and lower effective temperatures (so towards down right in the diagram). Massive stars can lose part of their initial mass by stellar winds.

Evolution Post Main Sequence

Evolution post the initial phase depends on the stellar mass. Suppose the initial mass of the star is M.

- For $M \leq 0.7 M_\odot$

  The initial hydrogen burning phase[4] of such stars is longer than the (estimated) age of the universe[5]. These stars still lie on the main sequence.

- For $M \leq 2.5 M_\odot$

  Central hydrogen burning is followed by a relatively brief phase of the hydrogen fusing into helium in a shell outside the center of the star. After this phase, the star moves

[4] Hydrogen burning just refers to its fusion into helium. Similarly, helium burning refers to its fusion into carbon.

[5] The current estimate for the age of the universe is around 13.787 billion years.

right in the HRD (lower effective temperature but same luminosity) and thereby strongly expands (1.5). This causes the temperature and density in the center rise enough so that helium starts fusing into carbon.

This helium burning is very explosive for these stars, resulting in 'helium flashes', during which large amounts of stellar mass get ejected. After these flashes, a new stable equilibrium configuration is established with a helium burning shell formed besides the hydrogen burning shell. Expanding its radius, the star will evolve into a red giant or supergiant and move along the asymptotic giant branch (AGB) in the HRD.

- For $M \leq 8M_{\odot}$

  After the initial expansion due to the formation of the hydrogen burning shell, these stars have their cores hot and dense enough for 'quiet' helium formation. Once the core runs out of helium, a second shell source forms fusing helium. In this stage, the star becomes a red giant or supergiant. A large portion of stellar mass is ejected into the interstellar medium in the form of stellar winds.

  The configuration of the helium shell is unstable and it releases flashes (helium is burned) in the form of pulses. After some time, this leads to the ejection of the outer envelope[6] which then becomes visible as a *planetary nebula*.

  At this stage, since the star is shedding its outer envelope, its radius decreases drastically, leading to a very drastic increase in temperature (1.5), moving it left in the HRD. The fuel burning in the core is not affected, hence the luminosity stays mostly the same. However, once the fuel runs out, there is a decrease in the luminosity which leads to a small decrease in the temperature until it cools down to become a *white dwarf* with a mass of about $0.6M_{\odot}$ and a radius roughly the same as that of Earth.

- For $M \geq 8M_{\odot}$

  For these stars, the temperature and density at its center become so tremendous that carbon can also be fused.Subsequent stellar evolution leads to a core-collapse supernova, leading to the formation of a neutron star, or for the most massive ones ($\sim M \geq 25M_{\odot}$), a black hole.

[6] The outer envelope is everything apart from the star's core.

Chapter 2

The World of Galaxies

Ever since the Hubble Deep Field image, the whole of humanity has set their sight to discover what lies far beyond they could ever comprehend. When dealing with extragalactic astronomy, the most notable entities are arguably other galaxies, and this is a chapter all about them.

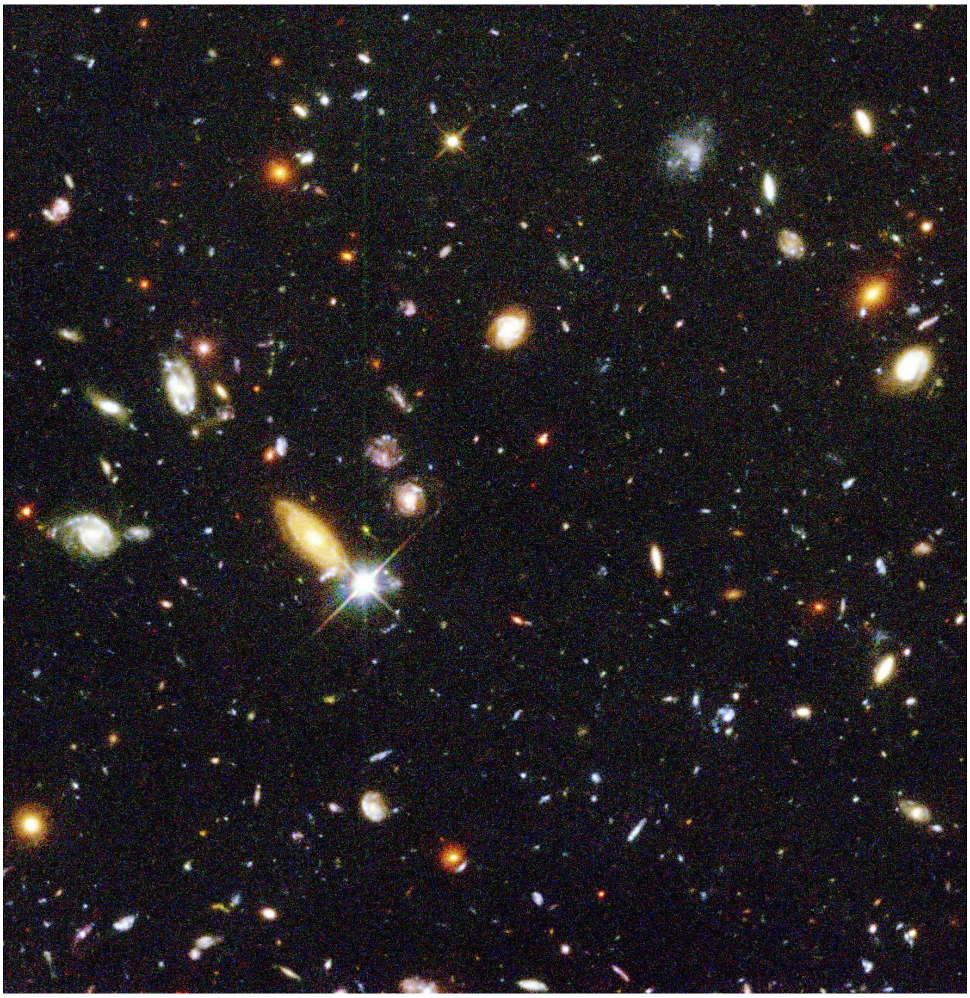


Figure 2.1: The Hubble Deep Field image produced by *Edwin Hubble* changed the world forever. All the dots in the image are galaxies, not stars.

The light from 'normal galaxies' is mainly the sum of all the light emitted by the stars present within it. As we learnt about stars, hot and blue stars have a much shorter lifespan than the cooler red supergiants or giants. Further, white dwarfs are far less luminous than red supergiants or giants, so that older groups of stars primarily look red.

2.1 Classification of Galaxies

2.1.1 Morphological Classification: Hubble Sequence

On the basis of shape or morphology, galaxies have been historically classified into the following categories:

1. **Elliptical Galaxies**: Their isophotes[1] resemble ellipses. They are further subclassified according to the **ellipticity** of the isophotes: $\epsilon = 1 - b/a$, where a, b are the semi-major and semi-minor axes. Then we define $N = 10\epsilon$ which gives us the representation EN for elliptical galaxies. An E7 galaxy is an elliptical galaxy with $b/a = 0.3$.
2. **Spiral Galaxies**: Their isophotes look like spirals. They are subdivided into two classes: *normal spirals* (S's) and *barred spirals* (SB's). These subclasses are further classified on the basis of the (decreasing) ratio of bulge to disc luminosity denoted by a, ab, b, bc, c, cd, d. There are also supposed correlations of bulge size (decreasing along the sequence) and spiral arm tightness (increasing along the sequence). The sequence goes from *early type* (Sa, SBab) to *late type spirals* (SBc, Sd). This is a purely historical convention.
3. **Irregular Galaxies** (Irr's): Galaxies with weak (Irr I) or no (Irr II) regular structure. This adds further subclasses to the Hubble sequence: Sdm, Sm, Im, Ir (m stands for Magellanic[2], Ir is just irregular).
4. **S0 Galaxies**: They are a transition between ellipticals and spirals, also called *lenticulars*. They contain a bulge and a large enveloping region of relatively unstructured brightness.

Early/Late Type Nomenclature

This is another purely historical convention with somewhat ambiguous definition that stuck along. It has nothing to do with evolutionary stages of the galaxies.

- **Ellipticals** and **S0 galaxies** are referred to as *early type galaxies*.
- **Spiral galaxies** (the remaining) are referred to as *late type galaxies*.

[1] Isophotes are contours along which the surface brightness is constant.

[2] Magellanic Clouds are two irregular dwarf galaxies orbiting the Milky Way. The term *Magellanic Clouds* was reported by the portuguese explorer Ferdinand Magellan's crew during their expedition.

Note that early-late type galaxies are distinct from early-late type spirals.

2.1.2 Spectral Distribution of Galaxies

In general, Elliptical galaxies appear red and spiral galaxies appear blue. Drawing inference from above, elliptical galaxies must be having a lot more older stars and lesser new star formation in comparison to spirals.

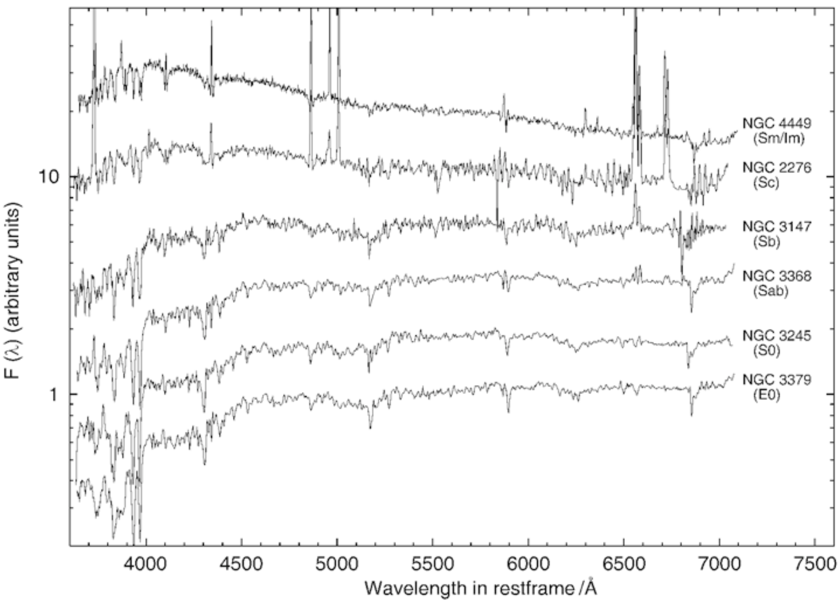


Figure 2.2: Spectra of different kinds of galaxies.

- **Normal Galaxies**: Galaxies whose total spectral distribution can roughly be described as the superposition of the spectra of all of its stars.
- **Active Galaxies**: The spectral distribution of these galaxies have major contribution from gravitational sources: specifically the infall of matter into the Supermassive Black Hole (SMB) at the center of the galaxy. *Active Galactic Nuclei* (AGNs) are a class of such galaxies in which the largest fraction of luminosity is produced in a very small central region: the active galactic nucleus.

2.1.3 Star Formation Rate

- Elliptical galaxies usually have marginal to no star formation.
- Spiral galaxies usually have star formation rates of around $\sim 3M_\odot$/yr.
- Starburst galaxies are those which have very high star formation rates $\sim 10M_\odot$/yr and more.

2.1.4 Colour of the Galaxies

Plotting the contour plot of the brighness of galaxies for different colours and absolute magnitude, we obtain the Colour-Magnitude Diagram (CMD):

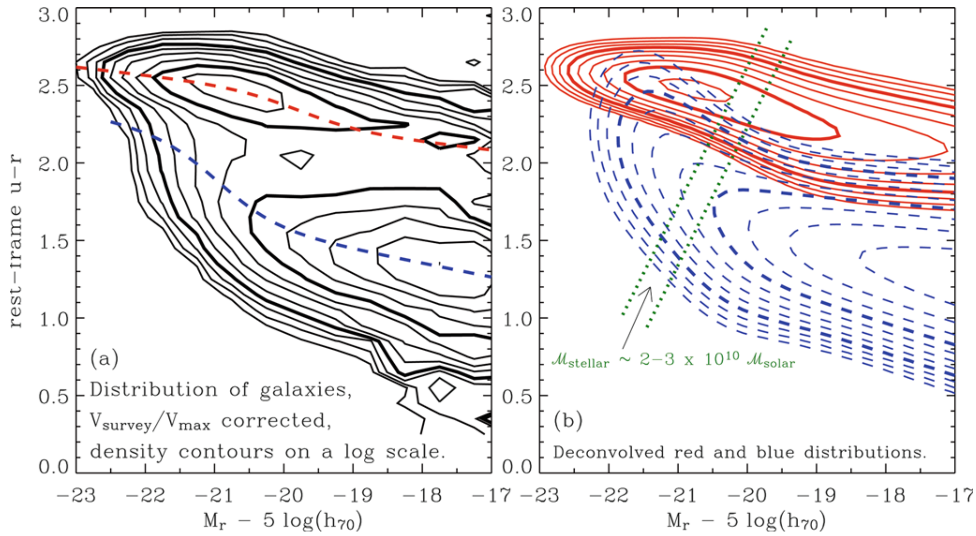


Figure 2.3: CMD showing two peaks of brightness, one at the redder and cooler portion and another at the bluer and hotter portion.

From this diagram we observe that most of the galaxies in the local group are either blue and hot or red and less hot. These two correspond to usual spiral and elliptical galaxies respectively. The local group can be described as a binodal distribution in colour with the characteristic colour having a slight dependence on the absolute magnitude.

A red stellar population does not contain massive luminous stars, leading to a high mass-to-light ratio. As a consequence, a red population of the same brightness would be much more significantly massive than a blue population.

2.2 Elliptical Galaxies

A rough subclassification of elliptical galaxies based on luminosity and size is as follows:

- *Normal Ellipticals*: Consists of giant ellipticals (gE's), those of intermediate luminosity (E's) and compact ellipticals (cE's). $M_{\rm B} \sim -23$ to -15.
- *Dwarf Ellipticals* (dE's): Different from cE's because they have a significantly lower surface brightness and a lower metallicity.
- *cD Galaxies*: Extremely luminous ($M_{\rm B} \sim -25$) and large (up to R $\lesssim$ 1 Mpc) galaxies found only at the center of dense clusters of galaxies. With very high surface brightness near the center, they have an extended diffuse envelope. Very high M/L ratio.
- *Blue Compact Dwaf Galaxies* (BCD's): They are very blue ($\langle B-V \rangle$ between 0.0 and 0.3) dwarf galaxies that contain an appreciable amount of gas.
- *Dwarf Spheroidals* (dSph's): Very low luminosity ($M_{\rm B} \sim -8$) and surface brightness.

Table 2.1: Characteristic values for early-type galaxies

	S0	cD	E	dE	dSph	BCD
M_B	-17 to -22	-22 to -25	-15 to -23	-13 to -19	-8 to -15	-14 to -17
M ($\mathrm{M}_\odot$)	10^{10}–10^{12}	10^{13}–10^{14}	10^{8}–10^{13}	10^{7}–10^{9}	10^{7}–10^{8}	$\sim 10^9$
D_{25} (kpc)	10–100	300–1000	1–200	1–10	0.1–0.5	< 3
$\langle M/L_B \rangle$	~ 10	> 100	10–100	1–10	5–100	0.1–10
$\langle S_N \rangle$	~ 5	~ 15	~ 5	4.8 ± 1.0	–	–

D_{25} denotes the diameter at which the surface brightness has decreased to 25 B-mag/arcsec^2, S_N is the "specific frequency", a measure for the number of globular clusters (x.) in relation to the visual luminosity, and M/L is the mass-to-light ratio in Solar units (the values of this table are taken from the book by Carroll & Ostlie).

2.2.1 Brightness Profile

The brightness profile of a galaxy usually follows the *de Vaucouleur's profile* which says the surface brightness measured as $\mu \propto -2.5\log(I)$ is proportional to the $r^{1/4}$ where r is the projected distance from the center. More specifically, we have:

$$\log\left(\frac{I(R)}{I_\mathrm{e}}\right) = -3.3307\left[\left(\frac{R}{R_\mathrm{e}}\right)^{1/4} - 1\right]$$

where R_e is the effective radius (v.) and I_e is the effective intensity (vi.). Another form of the above equation is:

$$I(R) = I_\mathrm{e}\exp\left(-7.669\left[(R/R_\mathrm{e})^{1/4} - 1\right]\right)$$

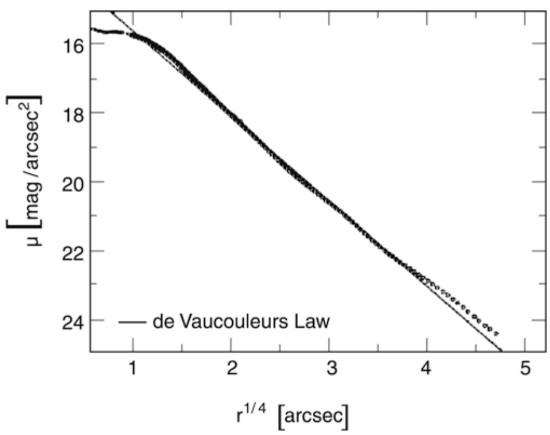


Figure 2.4: Surface brightness of galaxy NGC 4472 fitted with a de Vaucouleur's profile. It is evident that this galaxy has a light deficit; a core.

The brightness profiles of normal E's and cD's (in its innermost parts) approximately follows a de Vaucouleur's profile. The central brightness profile of ellipticals sometimes lies well below the $r^{1/4}$ fit. Such ellipicals are said to have a *core* or *light deficit*. These are usually very luminous, consequently also having very high stellar masses. The opposite can also happen with an *excess of light* in the central region of the galaxy, typically having lower luminosity and thus lower stellar mass.

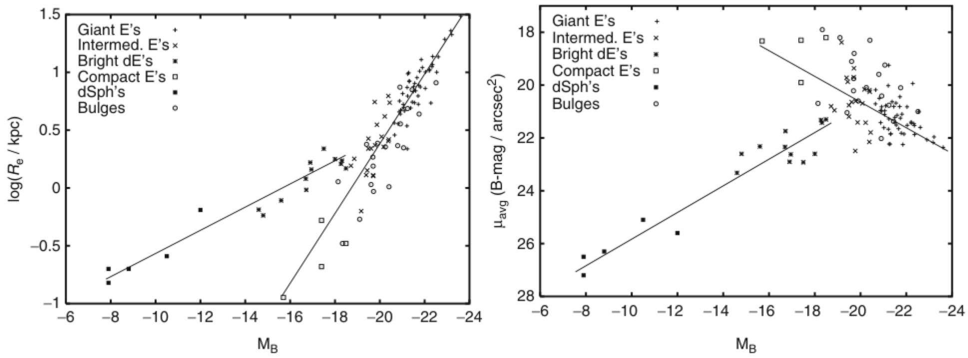


Figure 2.5: *Left*: effective radius R_{e} vs absolute magnitude M_{B}. *Right*: avg surface brightness μ_{avg} vs absolute magnitude M_{B}.

2.2.2 Chemical Composition

Many ellipticals (up to 75%) contain visible amounts of dust, manifested partially in the form of a dust disk; frequently extended disks of cold gas (HI), upto ∼200kpc in diameter.

Star Formation

Star formation is low but not existent. Around 15% of red galaxies showed a strong UV excess according to UV-observations from GALEX. Around 75% of them show clear signs of star formation. The UV emission (that signifies O and B stars primarily[3], which have a relatively short lifespan) is more extended than the optical image of the galaxies, indicating that the star formation occurs in the outer parts of the galaxy.

Metallicity

The metallicity of ellipticals and S0 galaxies increase towards the galaxy center, as derived from the colour gradients. S0 bulges also appear redder than the disk.

2.2.3 Dynamics of Elliptical Galaxies

We can tell from doppler measurements if a galaxy is rotating (unless it's directly face-on). We consequently try to understand how they are stable in time or in such shape and motion to begin.

Cause of Ellipticity

If rotation of the ellipticals were the only reason for their flattening from supposed 'spherical' galaxies, it can be shown through stellar dynamics that the rotational speed of the galaxy v_{rot} and velocity dispersion[4], σ_v has to obey:

$$\left(\frac{v_{\mathrm{rot}}}{\sigma_v}\right)_{\mathrm{iso}} = \sqrt{\frac{\epsilon}{1-\epsilon}}$$

[3]Out of all the spectral classes of stars. There are other sources of UV like AGNs.

[4]Measure of how much the stars are moving randomly in comparison to the mean velocity, just like gas molecules in random motion because of their thermal energy.

where ϵ is the ellipticity. While this could have been an explanation for ellipticals of low luminosity or bulges of disks, in general, for ellipticals with very high luminosity, $v_{\text{rot}} << \sigma_v$, so that rotation cannot be a major cause of the ellipticity.

Relaxation Time-scale

The brightness profile of an elliptical galaxy is determined by the distribution of its stellar orbits. We know from redshift measurements that galaxies rotate. Constructing initial position and velocity distributions that result in ellipticals not being stationary and being self-gravitating equilibrium systems is a difficult mathematical problem in general.

Assuming that close interactions of stars could lead to significant disruption in stellar orbits, *section 3.2.4* of *Schneider* ADD REFERENCE explains how the relaxation time turns out to be longer than the age of the universe, considering all stars of equal masses and extents.

We thus conclude that stars behave like a collision-less gas: they are stabilized by dynamical pressure, and they are elliptical because the stellar distribution is anisotropic in velocity space. This corresponds to an anisotropic pressure.

NEED TO EXPLAIN WHY LATER

2.3 Spiral Galaxies

Some trends are observed in the sequence of spiral galaxies, useful for early-late spiral classification:

- decreasing luminosity ratio of with $L_{\text{bulge}}/L_{\text{disc}} \sim 0.5$ for Sa's and ~ 0.03 for Sc's,
- increasing opening angle for the spiral arms, from $\sim 6°$ for Sa's to $\sim 18°$for Sc's,
- and increasing structure of the isophotes: Sa's have a 'smooth' distribution of stars along their spiral arms whereas Sc's have bright knots and HII regions.

Spirals also have lesser variation in their parameters of absolute magnitude and stellar mass in comparison to eliipticals. They range from $-16 \gtrsim M_{\text{B}} \gtrsim -23$ mag in terms of luminosity and $10^9 M_{\odot} \lesssim M \lesssim 10^{12} M_{\odot}$ in terms of stellar mass. Around 70% of all disk galaxies contain a large-scale stellar bar.

The morphology of spirals can be expanded according to the value of scale-height across the different stellar populations:

- **Young Thin Disk**: This component contains the largest fraction of gas and dust. Star formation is active and thee youngest stars can be found on this thin disk. Scale-height is the smallest.
- **Old Thin Disk**: This component is thicker and contains older stars.

- **Thick Disk**: This component is significantly thicker and consists of older, metal-poor stars with a higher velocity dispersion perpendicular to the disk plane. Scale-height is the largest.

Note that the scale-height corresponds to the age of the stellar population.

Table 2.2: Characteristic values for spiral galaxies

	Sa	Sb	Sc	Sd/Sm	Im/Ir
M_B	-17 to -23	-17 to -23	-16 to -22	-15 to -20	-13 to -18
M ($\mathrm{M}_\odot$)	10^9–10^{12}	10^9–10^{12}	10^9–10^{12}	10^8–10^{10}	10^8–10^{10}
$\langle L_{\mathrm{bulge}}/L_{\mathrm{tot}}\rangle_B$	0.3	0.13	0.05	–	–
Diam. (D_{25}, kpc)	5–100	5–100	5–100	0.5–50	0.5–50
$\langle M/L_B\rangle$ ($\mathrm{M}_\odot/L_\odot$)	6.2 ± 0.6	4.5 ± 0.4	2.6 ± 0.2	~ 1	~ 1
V_{max} range (km s^{-1})	163–367	144–330	99–304	–	50–70
Opening angle	$\sim 6°$	$\sim 12°$	$\sim 18°$	–	–
$\mu_{0,B}$ (mag arcsec^{-2})	21.52 ± 0.39	21.52 ± 0.39	21.52 ± 0.39	22.61 ± 0.47	22.61 ± 0.47
$\langle B - V\rangle$	0.75	0.64	0.52	0.47	0.37
$\langle M_{\mathrm{gas}}/M_{\mathrm{tot}}\rangle$	0.04	0.08	0.16	0.25 (Scd)	–
$\langle M_{H_2}/M_{HI}\rangle$	2.2 ± 0.6 (Sab)	1.8 ± 0.3	0.73 ± 0.13	0.19 ± 0.10	–
$\langle S_N\rangle$	1.2 ± 0.2	1.2 ± 0.2	0.5 ± 0.2	0.5 ± 0.2	–

V_{max} is the maximum rotation velocity, thus characterizing the flat part of the rotation curve. The opening angle is the angle under which the spiral arms branch off, i.e., the angle between the tangent to the spiral arms and the circle around the center of the galaxy running through this tangential point. S_N is the specific abundance of globular clusters as defined in (3.18).

2.3.1 Brightness Profile

The brightness profile of the bulge follows the de Vaucouleur's profile to a first approximation. The disk (face-on) follows an exponential profile. Expressing these distributions of the surface brightness $\mu \propto -2.5\log(I)$, we have:

$$\mu_{\mathrm{bulge}}(R) = \mu_{\mathrm{e}} + 8.3268\left[\left(\frac{R}{R_{\mathrm{e}}}\right)^{1/4} - 1\right] \tag{2.1}$$

and

$$\mu_{\mathrm{disk}}(R) = \mu_0 + 1.09\left(\frac{R}{h_R}\right) \tag{2.2}$$

where μ_{e} is the surface brightness at R_{e}, h_R is the scale-length (i.) and μ_0 is the central surface brightness of the disk component, which is measured by fitting the sum of the above equations to the total light profile of the galaxy.

The brightness profile of the edge of the spirals is exponential. The scale-height h_z (ii.) is almost independent from the galacto-centric radius[5] R and scales roughly with the rotational

[5]Distance of an object from the center of the galaxy.

velocity of the disk $v_{\rm rot}$.

$$\mu_{\rm edge}(z) = \mu_0 + 1.09\left(\frac{|z|}{h_z}\right) \tag{2.3}$$

The typical value of the ratio of scale-height to scale-length is $h_R/h_z \sim 0.07$.

Bulges and Pseudobulges

Classical bulges follow the de Vaucouleur's profile. However, there exist bulges which don't follow the de Vaucouleur's profile and instead follow a more exponential profile and are flatter. These bulges are called *pseudobulges*. Classical bulges lie on the same sequence as ellipticals on the effective radius vs absolute magnitude graph (Fig. 2.5), however the pseudobulges have lower luminosity for the same effective radius. Classical bulges ccontain about 10 times more stars than pseudobulges.

Some late type spirals have no bulge at the center but instead a nuclear stellar cluster, whose luminosity is dominated by a relatively young stellar population but whose stellar mass is dominated by an old population.

Freeman's Law

Freeman's law says that for 'normal spirals', the variation of central surface brightness μ_0 is very less across different galaxies. For Sa to Sc's[6], $\mu_0 = 21.52 \pm 0.39$ B-mag/arcsec^2 and for Sd and later types, $\mu_0 = 22.61 \pm 0.47$ B-mag/arcsec^2.

This however does not apply to *low surface brightness galaxies (LSBs)*. These galaxies have μ_0 2 or more orders lesser in magnitude, making them very hard to detect at larger distances. They have much slower star formation rates.

Warps in Disks

The disk warps out of the plane of the galactic disk where most stars and gas rotate. In most cases, the warps start at radii beyond the optical radius of the galaxy, hence only visible in the distribution and motion of gas (primarily neutral hydrogen atom gas HI).

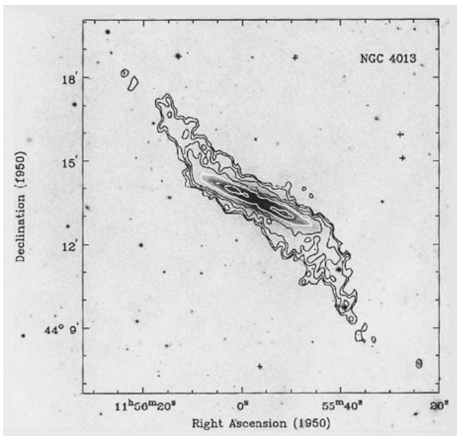


Figure 2.6: Edge-on spiral NGC4013, optical emission in grey, superimposed by intensity of cold 21-cm emitting gas (HI). The warping begins only about when the visible disk ends.

Stellar Halo

Stellar halos of distant galaxies is hard to study due to their low surface brightness. In our neighbouring galaxy M31, the Andromeda galaxy, a stellar halo of red giant branch (RGB) stars was detected which extends out to more than 150 kpc. For radii $r \lesssim 20$ kpc, it follows the extrapolation from the bulge surface brightness, but for

[6]B-mag refers to the magnitude measured in Johnson blue band.

larger radii, it follows a radial density profile of roughly
$\rho \sim r^{-3}$.

Thick Disk

This component (2.3) can only be studied for edge-on spirals. The scale height of a stellar population increases with its age. For luminous disk galaxies, the thick disk does not have any substantial contribution to the luminosity. However, for lower-mass disk galaxies with rotational velocities $\lesssim$ 120 km/s, the thick disk stars contribute nearly half of their luminosity and may dominate the stellar mass. This results in the dominant stellar population of these galaxies being old despite it appearing blue.

Sizes of Disks

Typically, the optical radius of a spiral galaxy extends out to about $4h_R$ (break radius), after which the surface brightness, and thus the stellar surface density, displays a break. The characteristic surface brightness at which this occurs is $\mu_{\mathrm{B}} \approx 25.5$ mag/arcsec^2. Although there are many exceptions to this behavior, it still prevails in the majority of spirals. In contrast to the stellar distribution, neutral gas is observed to considerably larger radius, typically upto twice the break radius.

2.3.2 Chemical Composition

Star Formation

Stars form from gas only where its surface mass density, Σ_{gas} (measured in $M_{\odot}\,\mathrm{pc}^{-2}$), exceeds a certain value. Marteen Schmidt in 1959 proposed its relation to the star formation rate per unit area, Σ_{SFR} (measured in $M_{\odot}\,\mathrm{yr}^{-1}\mathrm{kpc}^{-2}$) as:

$$\Sigma_{\mathrm{SFR}} \propto \Sigma_{\mathrm{gas}}^{N}$$

with power law index of $N \approx 1.4$. Detailed later by Rob Kennicutt, the new Schmidt-Kennicutt law now stands at:

$$\frac{\Sigma_{\mathrm{SFR}}}{M_{\odot}\,\mathrm{yr}^{-1}\mathrm{kpc}^{-2}} = (2.5 \pm 0.7) \times 10^{-4} \left(\frac{\Sigma_{\mathrm{gas}}}{M_{\odot}\,\mathrm{pc}^{-2}} \right)^{1.40 \pm 0.15}$$

Because of the apparent absence of star formation in the outer parts of spiral galaxies, this relation is complemented with a cutoff Σ_{gas}.

This graph shows data of star formation rates against availability of hydrogen for normal spirals, LSBs and starburst galaxies. Globally, the graph can be fitted by a power law of $N = 1.4$. However, for LSBs and normal

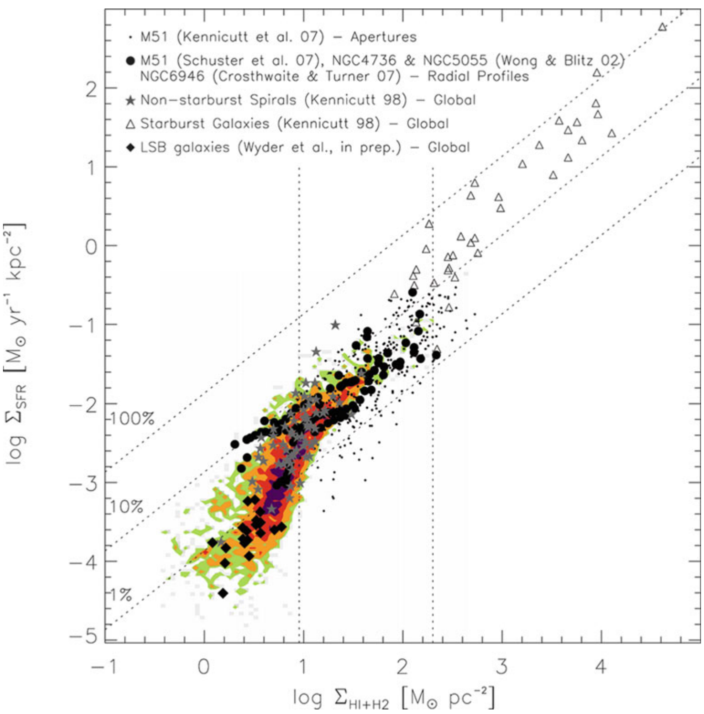


Figure 2.7: The diagonal lines show how long it will take to deplete available gas reservoir, corresponding to time scales of $10^8, 10^9$ and$10^1$0 from top to bottom.